

HIGH-PRESSURE GAS TRANSMISSION - INTERNAL FLOW COATING QUALITY PLAN (ITP)

Pipe & Coating Description: 36" (914.4 mm) x 14.30 mm, Grade X65M PSL 2 with Internal Flow Efficiency Liquid Epoxy Coating | **Governing Codes:** API RP 5L2 4th Ed. / ISO 15741 / BOTA / Shell DEP 31.40.20.37

No	Inspection / Test Stage	Frequency & Sampling	Location & Specimen	Test Standard & Procedure	Specified Acceptance Criteria & Tolerances	Governing Clause	Hold / Witness
1	Liquid Epoxy Material Verification	Every incoming batch of base & curing agent	Manufacturer QC batch certificate	API RP 5L2 / ISO 15741	Two-pack polyamide/amine cured epoxy. Non-volatile solids >= 60% by vol, pot life >= 2 hrs, flash point > 25 °C, EN 10204 3.1 cert	API RP 5L2 Cl. 2	H/R
2	Internal Surface Pre-Cleaning	100% pipe internal bore	Full inside surface	SSPC-SP 1 Solvent Wipe	Internal bore free of rust lumps, grease, lubricants, protective lacquers; washed with clean solvent or alkaline wash	ISO 15741 Cl. 6.1	W
3	Internal Abrasive Blast Cleaning	100% of all pipes (automated lance)	Full internal pipe wall	ISO 8501-1 / SSPC-SP 10	Cleanliness: Sa 2.5 (Near-White Metal). Free from all mill scale, oxides, and rust stains; uniform matte finish	API RP 5L2 Cl. 3.1	H/W
4	Internal Surface Roughness & Profile	At start of shift and every 2 hours	Replica Tape (Press-O-Film)	ISO 8503-2 / ASTM D4417	Anchor pattern profile: Rz 25 µm - 50 µm (target 35 µm). Sharp profile without rolled-over steel peaks	ISO 15741 Cl. 6.2	W
5	Internal Dust & Soluble Salt Removal	100% pipes air blow + 2 Bresle tests/shift	Internal bore & pipe ends	ISO 8502-3 / ISO 8502-6/9	Dust level Class 1 max (vacuum cleaned); Residual soluble salts <= 20 mg/m² (conductance check)	ISO 15741 Cl. 6.3	W
6	Environmental Application Conditions	Continuous logging during application	Coating bay environment	Digital Psychrometer / Dew Check	Substrate temp >= 10 °C and MINIMUM 3 °C above dew point; Relative Humidity (RH) <= 80%; ambient temp 10-40 °C	API RP 5L2 Cl. 3.2	W
7	Automated Internal Spray Application	100% of all pipes (automated lance boom)	Full internal length	Airless Spray System	Uniform travel speed and rotation; wet film thickness (WFT) monitored continuously during application pass	API RP 5L2 Cl. 3.3	W
8	Dry Film Thickness (DFT) Verification	100% of all pipes (12 points per pipe)	Clock positions at ends and center	ISO 2808 / SSPC-PA 2	Dry film thickness (DFT): Nominal 50 µm - 75 µm (minimum 40 µm, maximum 100 µm). Zero runs, sags, or blisters	API RP 5L2 Cl. 3.4	H/W
9	Surface Roughness of Cured Coating	1 pipe per 50 pipes (or 1 per shift)	Internal coated surface	Stylus Profilometer (ISO 4287)	Cured coating surface roughness: Rz <= 25 µm, Ra <= 3.0 µm (guaranteeing >= 5% gas throughput increase)	ISO 15741 Cl. 7.2	H/W
10	Degree of Cure (Solvent Rub Test)	1 pipe per shift / test unit	Cured internal coating surface	ASTM D4752 / ISO 15741 Annex C	50 double rubs with MEK (Methyl Ethyl Ketone) soaked rag; no coating softening, stickiness, or pigment removal	ISO 15741 Annex C	H/W
11	Cross-Hatch Adhesion Test	1 test per shift (on cutback ring)	Internal coated test coupon	ISO 2409 / ASTM D3359 Method B	Cross-cut adhesion rating: Class 0 (100% lattice squares completely intact, no flaking along incisions)	API RP 5L2 Cl. 4.2	H/W
12	Resistance to Gas Decompression (Blistering)	Qualification test per paint system	Pressure autoclave laboratory cell	ISO 15741 Annex D	Decompression from 100 bar natural gas/nitrogen at 10 bar/min; 10 cycles; ZERO blistering or loss of adhesion	ISO 15741 Annex D	H/R
13	Internal Cutback & Bevel Protection	100% both ends of each pipe	Both internal pipe ends	Steel rule / masking gauge	Internal coating cutback: 50 mm +/- 10 mm from pipe face for field girth welding; edges feathered smoothly	API RP 5L2 Cl. 3.5	W
14	Internal Holiday & Pinhole Inspection	Spot check 1 pipe per 10 pipes (or 100%)	Full internal coated length	Low-voltage sponge tester (67.5 V)	Low voltage wet-sponge detector; zero continuous pinholes or conductive defects through to steel substrate	NACE SP0188	W
15	Internal Cleanliness & Cap Installation	100% of finished line pipes	Internal bore & bevels	Visual / End cap verification	Internal bore blown completely clean and dry; UV-stabilized heavy duty polyethylene bevel caps strapped securely	API RP 5L2 Cl. 5	W